



Working with Labeled Volumes

A labeling session generates a `groundTruthMedical` Object and labeled volume data. These can be viewed and analyzed outside the app.

Navigate to the Module 3 folder in the course files and work through the **AnalyzingLabeledVolumes.mlx** script. You'll work with a completed labeled volume of the MRI data you saw in the previous video to compute properties like volume for each label.

We encourage you to work through the code in MATLAB. A static file is provided below. The code may be different than in MATLAB because interactive tools are not present in static files.

< 1 / 4 >
🔍 100% 🔍
📄 🔄

Analyzing Labeled Volumes

Table of Contents

- Overview.....1
- Inspect gTruthMed Object.....1
- Inspect Labeled Volume.....2
- View Labeled Volume.....3
- Measure Region Properties.....3
- Measure Labeled Voxel Values.....4
- Hounsfield Units.....4

Overview

When labeling a volume, the Medical Image Labeler App first creates a new Label Data volume the same size as the original volume and assigns all the voxels a value of 0. As labels are drawn in the app, the corresponding voxel values in the labeled volume are set equal to the label ID.

The `groundTruthMedical` object created by the labeling session ties all the session data together. It contains the following information:

1. `dataSource`: The path to the original medical images for each volume
2. `labelData`: The path to the corresponding labeled volumes
3. `labelDefinitions`: A table containing the name, color, and ID of each label

gTruthMed
1x1 groundTruthMedical

Property	Value
<code>DataSource</code>	<code>1x1 VolumeSource</code>
<code>LabelData</code>	<code>*C:\Medical Image Analysis\Labeled ...</code>
<code>LabelDefinitions</code>	<code>2x3 table</code>

Inspect gTruthMed Object

The label session folder and `grouthtruthmed.mat` file are created at the same time. Initially, the property values are empty. They are updated as the corresponding data is added to the session.

The code in the next section displays the property values of a `groundTruthMedical` object you specify. If you have not yet done so, follow these steps to create a `grouthtruthmed.mat` file:

1. Start a volume session in the Medical Image Labeler App
2. Import one of the MRI data sets provided in the course files
3. Define a label
4. Add a label to your volume